Continuous Agent Trajectory Stability Demo
Project Paper

Stability Oracle Demo Update

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1 Purpose

This paper records the second public demonstration domain for Stability Oracle: continuous 2-D agent trajectories.

The purpose is narrow.

> This demo shows that the oracle can classify instability in a continuous motion domain using the same core engine as the LLM reasoning demo.

This is not a physics experiment, robotics stack, or navigation benchmark. It is a deterministic trajectory-stability demonstration.

2 Frozen Demo Contract

Demo v1 is intentionally frozen. The current contract is:

- dataset size = 15
- telemetry dimension = 8
- seeded deterministic simulation
- expected bands: stable → stable, marginal → marginal, unstable → unstable
- oracle kernel unchanged

This prevents silent drift while the demo serves as a public proof surface for domain decoupling.

3 Dynamics

The demo uses a damped goal-seeking point agent in two dimensions.

State variables:

- position: $p_t = (x_t, y_t)$
- velocity: $v_t = (v_{x,t}, v_{y,t})$
- goal: $g = (g_x, g_y)$

Update rule:

$$a_t = k_p(g - p_t) - k_d v_t + \eta_t + b_t$$

$$v_{t+1} = v_t + dt \cdot a_t$$

$$p_{t+1} = p_t + dt \cdot v_{t+1}$$

The noise term η_t is always seeded and deterministic. The bias term b_t is either zero, sinusoidal drift, or rotational bias depending on the regime family.

4 Regime Families

The dataset is built from three fixed parameter families:

- stable: strong damping, low noise
- marginal: lower damping, sinusoidal bias, moderate noise
- unstable: weak damping, rotational bias, higher noise

Each family contributes five trajectories, differing only by seed.

5 Telemetry Definition

Each simulation frame is converted into a fixed 8-dimensional telemetry vector:

1. x position
2. y position
3. x velocity
4. y velocity
5. heading angle
6. curvature estimate
7. distance to goal
8. control signal magnitude

All values are numeric, fixed-width, and deterministically normalized. No semantic extras are placed inside the state vector.

6 Oracle Reuse Path

The oracle kernel is not modified. The demo path is:

```
trajectory
-> AgentTrajectoryAdapter
-> TelemetrySequence
-> existing normalizer
-> existing geometry encoder
-> existing telemetry bridge
-> existing OracleEngine
```

This is the main result of the demo. The second domain reuses the existing logical oracle instead of reimplementing a new classifier.

7 Dataset

The dataset contains fifteen deterministic trajectories:

- 5 stable
- 5 marginal
- 5 unstable

Each record stores:

- trajectory identifier
- regime label
- seed
- parameter bundle
- raw per-step dynamics values

This makes the demo fully inspectable and reproducible.

8 Observed Results

The current demo run produces clean band separation:

- stable trajectories $\rightarrow$ 5 stable
- marginal trajectories $\rightarrow$ 5 marginal
- unstable trajectories $\rightarrow$ 5 unstable

Representative metric behavior is also ordered by regime:

- stable trajectories retain full structure with near-perfect temporal consistency
- marginal trajectories keep most structure but show higher deformation and reduced geometric integrity
- unstable trajectories lose structure, show compressed temporal recovery, and degrade geometric integrity sharply

9 Interpretation

This domain is intentionally maximally different from the first public demo:

- the LLM reasoning demo is symbolic and discrete
- this demo is continuous and geometric

The point is not realism. The point is that the same oracle can classify structural quality in both settings using the same kernel and policy surface.

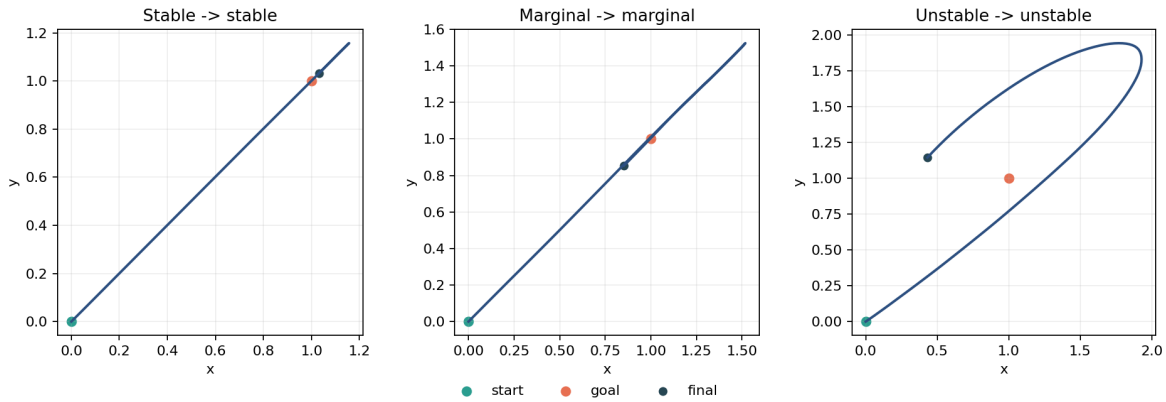

Figure 1: Representative stable, marginal, and unstable agent trajectories with oracle verdicts.

10 What This Demo Does Not Claim

This demo does not claim:

- robotics competence
- physics realism
- control optimality
- planning intelligence
- universal motion safety

It only demonstrates that the Stability Oracle can classify trajectory stability in a continuous motion domain through the same reusable engine path.

11 Operational Outputs

The current demo generates:

- `results.csv`
- `trajectory_plot.png`
- `DEMO_V1_FROZEN.md`
- deterministic regression tests for dataset generation and regime band separation

12 Conclusion

This is the correct scale for the second public domain.

The demo is:

- small
- deterministic
- visually obvious
- reusable through the same oracle kernel

- explicit about what it does and does not test

Most importantly, it makes the decoupling claim more concrete: the oracle is not limited to symbolic traces and can operate on continuous trajectory telemetry without changing its core semantics.